



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Pediatric Pulmonary Disease presentation and treatment Parts I and II

Course: PPOM II

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Session Objectives

1. Understand the epidemiology, pathophysiology, and clinical presentation of **cystic fibrosis**, **asthma**, **pneumonia** and **bronchiolitis**
2. Distinguish between multiple causes of respiratory distress in newborns, **including transient tachypnea of the newborn** and **meconium aspiration syndrome**
3. Identify radiographic differences between common pediatric respiratory pathologies
4. Distinguish between multiple causes of pediatric upper airway disease, **including epiglottitis**, **retropharyngeal abscesses**, and **peritonsillar abscesses**
5. Understand the social factors in pediatric pulmonology, including environmental health and vaccine non-adherence

Source: Course Syllabus

Case 1

A 15-month-old toddler, presents to your general pediatrics office for the first time with a wet cough for a “long time”.

His mother reports a history of recurrent pneumonia, one episode of which required hospitalization.

She also reports that he seems small for his age. He sometimes has bowel movements that smell really bad.

The patient was adopted. Birth history: unremarkable, biological parents: unknown

He lives with his adoptive parents in a home built in 2002.

Case 1: Physical Exam

- **General:** Ill-appearing, frequent moist cough, small for age
- **VS:** T 100.5F, HR 166, RR 45, Ht: 20%, Wt: 10%, HC: 20%
- **Cardio:** pos. S1, S2, no MRG
- **Resp:** pursed lips, wheeze and rhonchi in all lung fields

- Chest X-Ray:



Case 1: What is the diagnosis?

- A. Asthma
- B. Cystic Fibrosis
- C. Foreign Body aspiration
- D. Bronchiolitis
- E. Laryngotracheobronchitis

Case 1: What is the diagnosis?

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B. Cystic Fibrosis

C. Foreign Body Aspiration

D. Bronchiolitis

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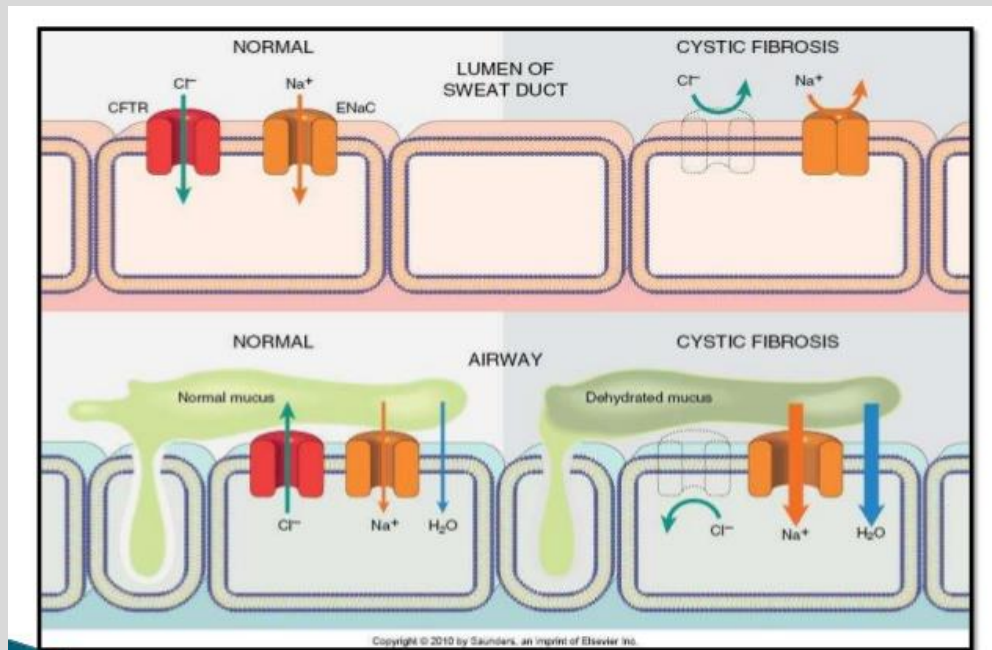
Cystic Fibrosis: Epidemiology

- Autosomal recessive, life-shortening
- >70,000 affected worldwide
- Incidence: predominantly European descent but all races/ethnicities can be affected
- Involves the long arm of chromosome 7 which encodes for the Cystic Fibrosis Transmembrane Regulator(CFTR): chloride channel on apical surface of epithelial cells(water and salt across membranes)
 - Most common mutation(almost 2000) is deletion of 3 base pairs at the 508($\Delta F508$) position resulting in absence of phenylalanine

Pathogenesis

- Epithelial cells in the **airways** have reduced ability to secrete Cl^-
- causing airway epithelium to reabsorb excessive sodium and therefore water, causing dehydration of airway secretions
 - Failure to clear mucus secretions leads to endobronchial colonization(esp. *S. aureus*/*P. aeruginosa*)

Salty sweat: Chloride cannot be reabsorbed. Sodium follows chloride to remain in the lumen of sweat ducts



Clinical Signs and Symptoms

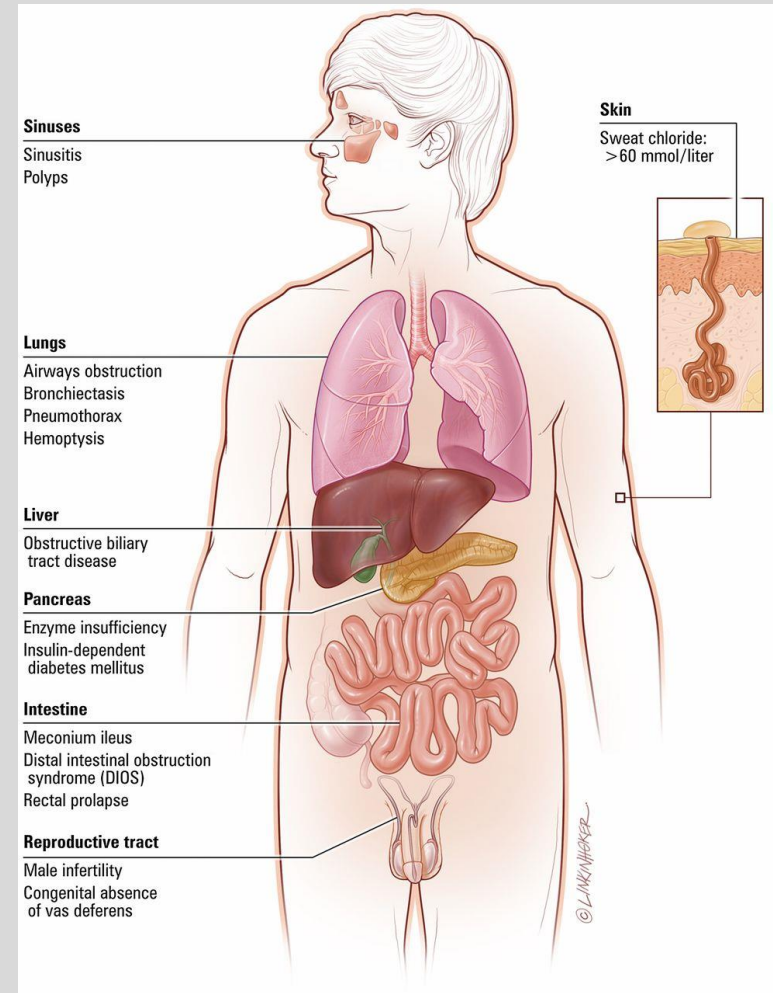
- Neonatal/GI:
protein/sugars(less so)
and esp. fat
malabsorption, failure to
thrive, steatorrhea, liver
disease
- Pulmonary: chronic
cough, chronic resp.
infection, nasal polyps,
recurrent sinusitis,
pneumonia
- Other:
 - Vitamin deficiency (A,
D, E, K-fat soluble)
 - CF-related diabetes
 - Infertility
 - Electrolyte
disturbances
 - Digital clubbing
 - Congenital absence of
vas deferens

Case 1: Diagnosis

- Newborn screening: all 50 states
- **Sweat Cl testing(positive: >60meq or mmol/L)**
- Criteria:
 - ≥ 1 typical clinical feature AND
 - 2 elevated sweat Cl tests OR
 - 2 known CF mutations OR
 - abnormal nasal epithelial potential difference

Case 1: Some Indications for Sweat Cl testing

- Respiratory:
 - Chronic/recurrent cough
 - Recurrent bronchiolitis
 - Recurrent wheeze/difficult to control asthma
 - Unexplained respiratory *P. aeruginosa*
- Gastrointestinal:
 - Meconium ileus
 - Neonatal intestinal obstruction
 - Steatorrhea, malabsorption
- Miscellaneous:
 - Positive newborn screen
 - Digital clubbing
 - Failure to thrive
 - Family history



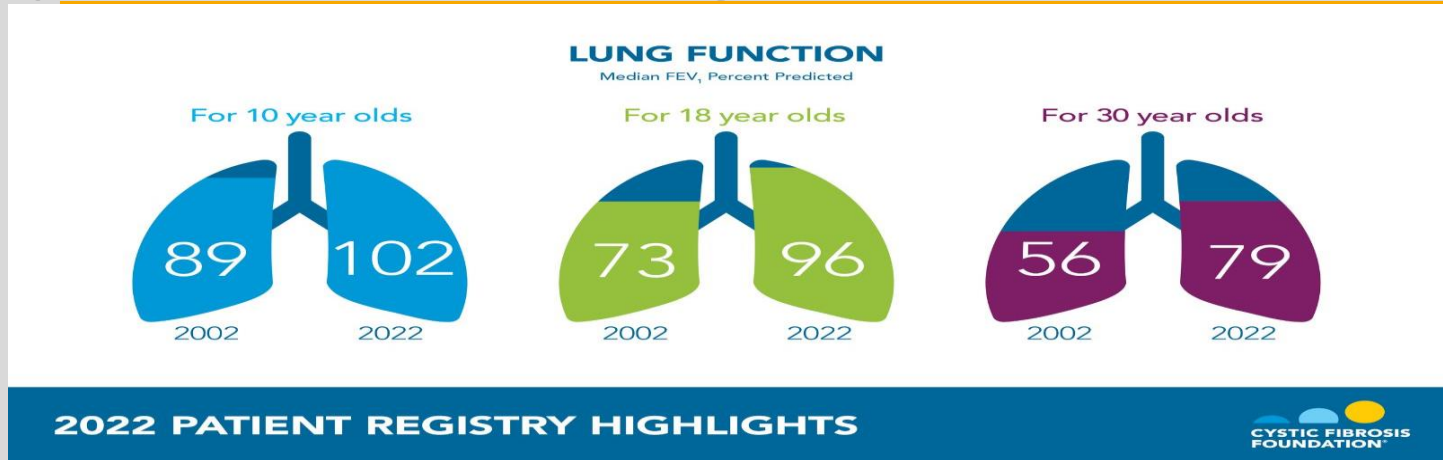
Text adapted from Nelson's Essentials of Pediatrics Table 137.2, 2023

Image: Pediatrics in Review May 2014, VOLUME 35 / ISSUE 5

Case 1: Management

- Respiratory: clearing airway secretions
 - Chest physiotherapy
 - OMM
 - Aerosolized DNase + saline (nebulized)
 - Infection: aggressive IV antibiotics
- Pancreatic insufficiency:
 - Enzyme capsules (lipase, protease)
 - Vitamin supplementation (A, D, E, K)
- Meconium Ileus: enemas
- CF-related diabetes: insulin
- Cholestasis: ursodeoxycholic acid
- Even well-controlled patients have a long problem list
- 2019 CFTR gene modulator approved by the FDA. For some has improved overall symptoms.

Cystic Fibrosis: Prognosis



- Do not restrict activity
- Increasing life expectancy, quality of life
- Multidisciplinary care in CF centers

Image: Cystic Fibrosis Foundation 2022

Case 2

A 9 yo child presents to the Emergency Department with his parent.

He complains of a 10-hour history of wheezing and shortness of breath.

His mother states he had “a bad cold” for the past 3-4 days.

He was given one albuterol nebulizer treatment at home an hour ago with no relief.

Case 2: Physical Exam

General: Ill-appearing, in distress, seated upright with hands on knees and leaning forward

Vitals: Temp 99F, HR 124, BP 128/94, RR 38, Pulse Oximetry: 90% on room air

Resp: sternocleidomastoids appear hypertonic bilaterally, wheeze appreciated on exhalation in all lung fields, and slight upper-lobe inspiratory crackles

Cardio: +S1/ S2 heard, no MRG

Case 2: What is the diagnosis?

- A. Bronchiolitis
- B. Pneumonia
- C. Asthma
- D. Allergic Rhinitis

Case 2: What is the diagnosis?

- A. Bronchiolitis
- B. Pneumonia
- C. Asthma
- D. Allergic Rhinitis

Asthma: Pathophysiology

- Episodic airway hyperreactivity due to:
 - Allergens
 - Irritants
 - Infection
 - Exercise
- Possible permanent airway remodeling: increased extracellular matrix(ECM) proteins, vascular hyperplasia leading to decreased pulmonary function

Asthma: Immunology

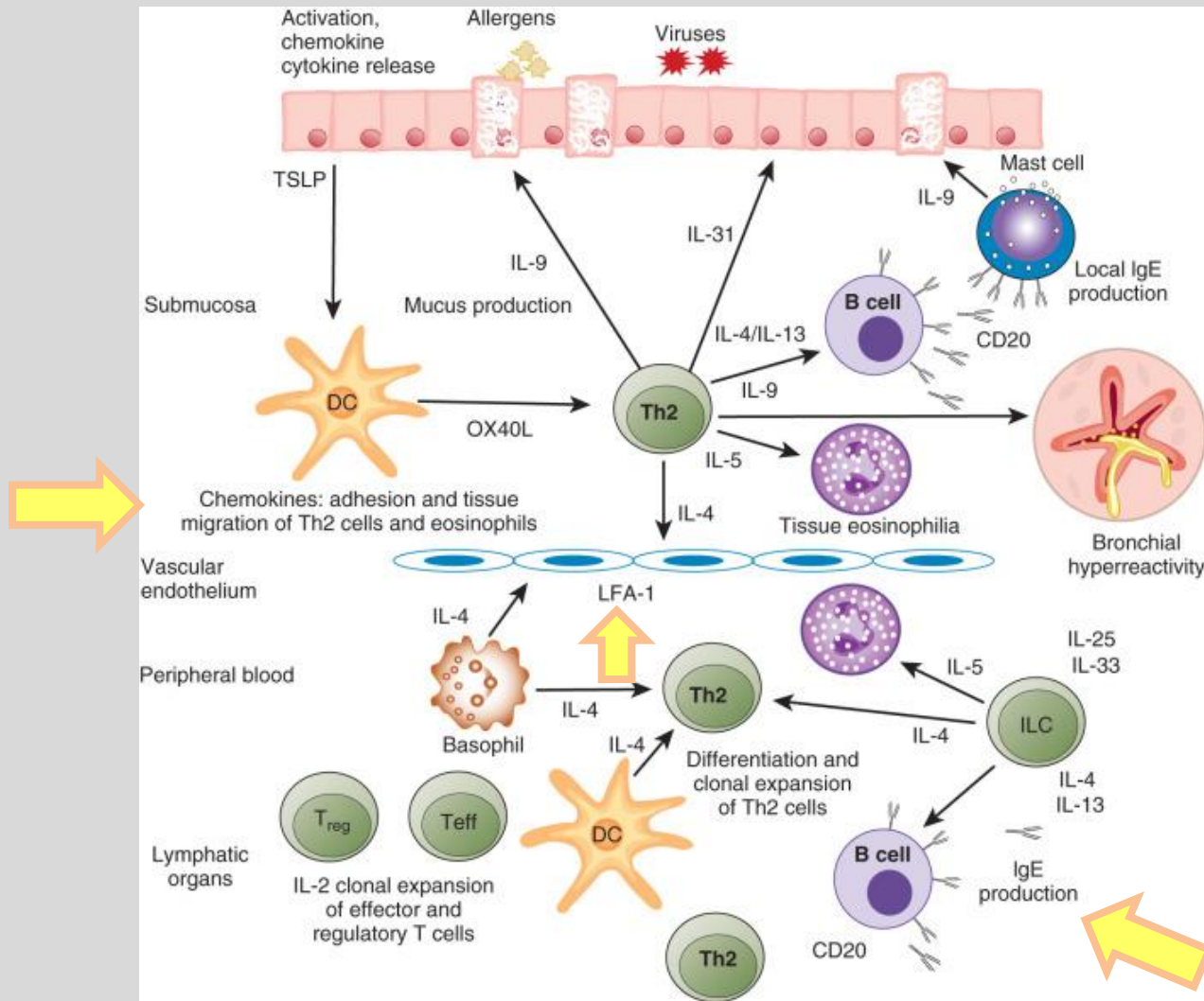


Image: Fig. 169.2 from Nelson's Textbook of Pediatrics 21e

Asthma: Epidemiology

- 4.7 million children(CDC) under 18(6.2% '21) in the U.S.
- Children: males > females (opposite in adulthood), approx. 8% overall population. Stable numbers but disparities exist.
- Past 20 years: **2-7x** more ED visits and hospitalizations (2017 1.6 mil ED and 183,000 hospitalizations). increased deaths in black children. ED/UCC visits higher in South and large metro areas.
- 2016-2018 CDC: death rates higher among adults, females and blacks- Northeast, Midwest and West. Good news for the South and young people. Higher in rural vs urban.
- Associations: small and medium metro areas, low income, Black, Puerto Rican, allergic conditions
- Strongest risk factors: atopy, family history

Asthma prevalence by race and ethnicity <18yrs of age(CDC '19-21):33% at or below poverty line

White NH 7.6 %

Black NH 10.9 %

AI/AN NH 12.3 %

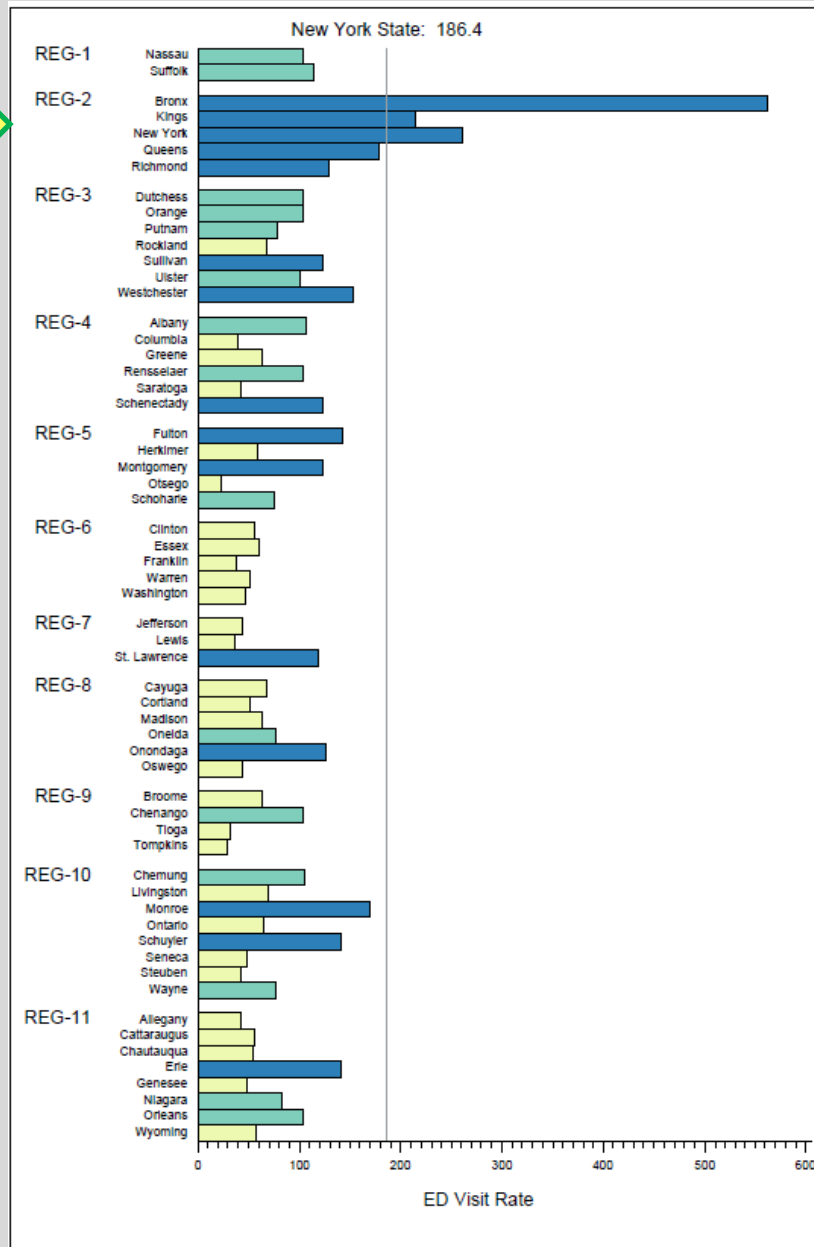
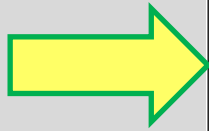
Asian NH 4.1 %

Hispanic 5.9 %

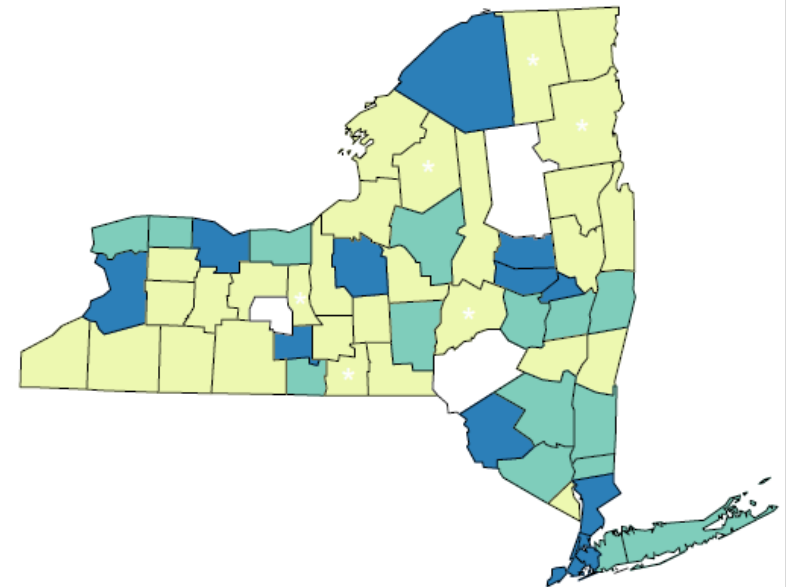
Abbreviations: NH = Non-Hispanic

AI/AN = American Indian/ Alaska Native

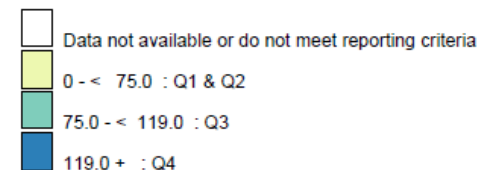
New York State



Asthma emergency department visit rate per 10,000
Aged 0-4 years
2016



ED Visit Rate
Counties Are Shaded Based On Quartile Distribution
(* Fewer than 10 events in the numerator, therefore the rate is unstable)



Source: 2016 SPARCS Data as of December 2017

Arkansas

Ann Allergy Asthma Immunol. 2010 February ; 104(2): 125–131. doi:10.1016/j.anai.2009.11.038.

A comparison of asthma prevalence and morbidity between rural and urban schoolchildren in Arkansas

Robbie D. Pesek, MD^{*}, Perla A. Vargas, PhD[†], Jill S. Halterman, MD, MPH[‡], Stacie M. Jones, MD^{*,§}, Andy McCracken, MS^{*}, and Tamara T. Perry, MD^{*,§}

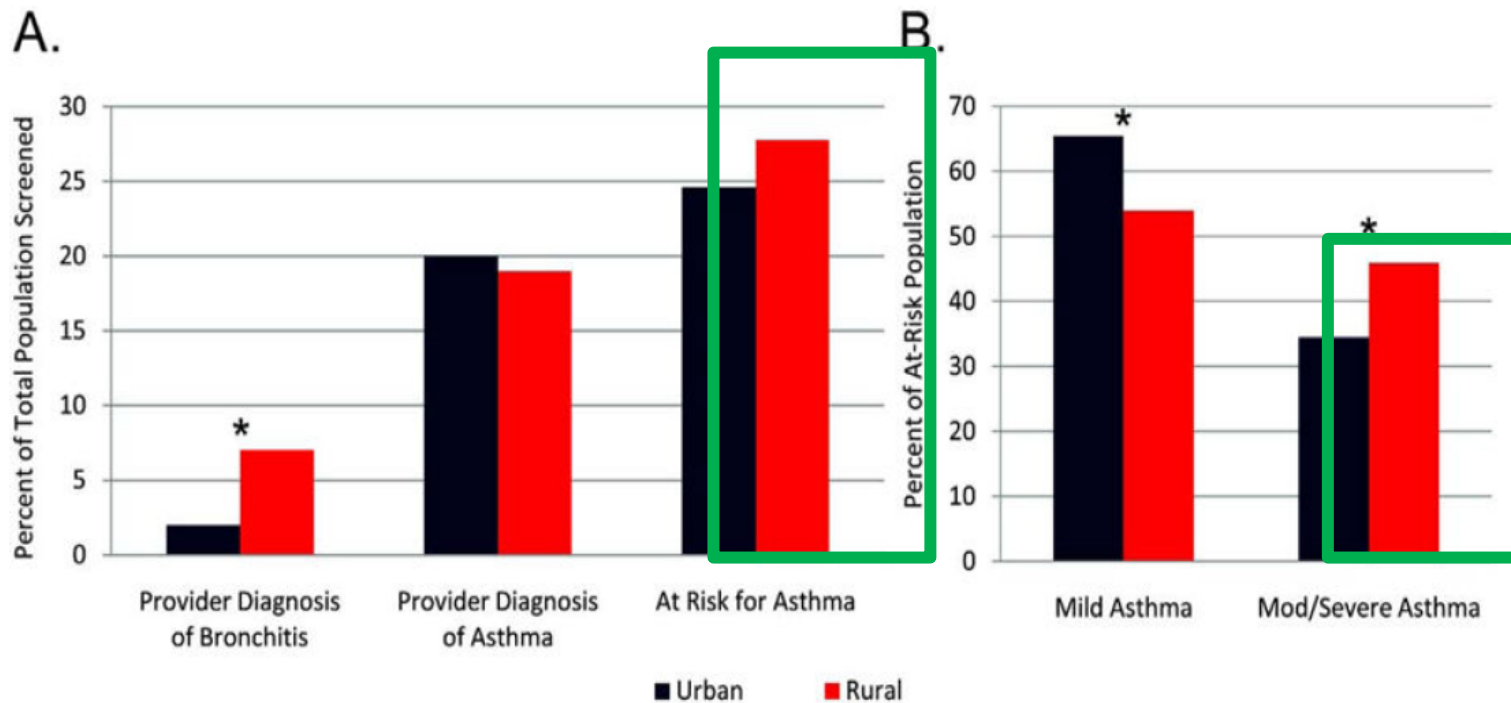


Figure 2.

A, Comparison of provider-diagnosed chronic bronchitis, provider-diagnosed asthma, and at-risk children in urban and rural settings. B, Asthma severity in at-risk urban and rural children. * $P < .001$.

Return to the Patient

Pt is administered a nebulized formulation of albuterol + ipratropium bromide. Three hours after initial presentation, repeat vitals:

T: 98.8F, HR: 90, RR: 20, SpO2: 98% (room air)

When you ask a more detailed history, pt. and his mom report that he has shortness of breath each day at recess or gym class, wakes up two nights a week wheezing, and “doesn’t always get to the end of the month” with each inhaler.

How would you classify our patient’s asthma?

Asthma Classification

Classification of Asthma Severity (5-11 years of age)

Components of Severity	Persistent			
	Intermittent	Mild	Moderate	Severe
Daytime symptoms	≤2 d/wk	>2 d/wk (but not daily)	Daily	Throughout the day
Nighttime awakenings	≤2×/month	3-4×/mo	>1×/wk (but not nightly)	Nightly
Short-acting β_2 -agonist use for symptom control (excludes using it for prevention of EIB)	≤2 d/wk	>2 d/wk (but not daily)	Daily	Several times per day
Interference with activity	None	Minor limitation	Some limitation	Extremely limited
Lung function	Normal FEV1 between exacerbations FEV1 >80% predicted FEV1/FVC >85%	FEV1 >80% of predicted FEV1/FVC >80%	FEV1 = 60%-80% of predicted FEV1/FVC = 75%-80%	FEV1 <60% of predicted FEV1/FVC <75%
Number of exacerbations requiring systemic steroids	0-1/y	≥2/y		
Therapy	Short-acting β -agonist (SABA) prn	Add low-dose inhaled corticosteroid (ICS)	Low-dose ICS + montelukast or medium-dose ICS	Refer to pulmonologist Medium-dose ICS + long-acting β -agonist

Date of download: 08/09/19 from Case Files Collection: casefiles.mhmedical.com, Copyright © McGraw-Hill Education. All rights reserved.

Case 2: DX/Management

- Diagnostic: spirometry(>5yr of age)
- Stepwise therapy, multiple drug classes
 - Inhaled corticosteroids (↓morbidity, may affect growth)
 - Leukotriene modifiers ([1] Leukotriene receptor antagonists, [2] leukotriene synthesis inhibitors)
 - Long acting beta-2 agonists (Formoterol, Salmeterol)
 - Theophylline, cromolyn sodium (rarely used)
 - Biologics, monoclonal antibodies (anti-IgE, anti-IL5, anti-IL5R, anti-IL4/IL13)
 - Short-acting beta-2 agonists (albuterol)
 - Anticholinergics (Ipratropium)
 - Oral(systemic) corticosteroids (avoid prolonged use)

Asthma Triggers

Trigger	Reduction Tactic
Viral upper resp. infection	↓Exposure; flu/COVID vaccine
Smoke (tobacco, wood)	Quit smoking (parents and caregivers); alternate heating
Dust mites	Bedding encasements and cleaning
Animal Dander	Air filters, separation from pets
Cockroach allergens	Pesticides, crack sealants
Indoor Mold	Cleaners, dehumidifiers

Adapted from Nelson's Essentials of Pediatrics Table 78.3

Asthma Management: Stepwise Approach

From Nelson's Essentials of
Pediatrics Figure 78.1, 2023

		Step up if needed (first check inhaler technique, adherence, environmental control, and comorbid conditions) Assess control Step down if possible (and asthma is well controlled at least 3 months)					
		Step 1	Step 2	Step 3	Step 4	Step 5	Step 6
Children 0–4 years of age		Intermittent asthma Persistent asthma: Daily medication Consult with asthma specialist if Step 3 care or higher is required. Consider consultation at Step 2.					
	Preferred	SABA PRN	Low-dose ICS	Medium-dose ICS	Medium-dose ICS + LABA or Montelukast	High-dose ICS + LABA or Montelukast	High-dose ICS + LABA or Montelukast + Oral corticosteroids ICS
	Alternative		Cromolyn or Montelukast				
	Quick-Relief Medication	Each Step: Patient Education and Environmental Control - SABA as needed for symptoms. Intensity of treatment depends on severity of symptoms. - With viral respiratory symptoms: SABA q 4–6 hours up to 24 hours (longer with physician consult). Consider short course of oral systemic corticosteroids if exacerbation is severe or patient has history of previous severe exacerbations. Caution: Frequent use of SABA may indicate the need to step up treatment. See text for recommendations on initiating daily long-term-control therapy.					
Children 5–11 Years of Age		Intermittent Asthma Persistent Asthma: Daily Medication Consult with asthma specialist if Step 4 care or higher is required. Consider consultation at Step 3.					
	Preferred	SABA PRN	Low-dose ICS	Low-dose ICS + LABA, LTRA, or Theophylline	Medium-dose ICS + LABA	High-dose ICS + LABA	High-dose ICS + LABA + Oral corticosteroids
	Alternative		Cromolyn LTRA Nedocromil, or Theophylline	Medium-dose ICS	Medium-dose ICS + LTRA or Theophylline	High-dose ICS + LTRA or Theophylline	High-dose ICS + LTRA or Theophylline + Oral corticosteroids
	Quick-Relief Medication	Each Step: Patient Education and Environmental Control, and Management of Comorbidities Steps 2–4: Consider subcutaneous allergen immunotherapy for patients who have persistent, allergic asthma. SABA as needed for symptoms. Intensity of treatment depends on severity of symptoms: up to 3 treatments at 20-minute intervals as needed. Short course of oral systemic corticosteroids may be needed. Caution: Increasing use of SABA or use >2 days a week for symptom relief (not prevention of EIB) generally indicates inadequate control and the need to step up treatment.					

Case 2: Asthma Action Plan

Asthma Action Plan

For: _____ Doctor: _____ Date: _____
 Doctor's Phone Number _____ Hospital/Emergency Department Phone Number _____

GREEN ZONE

Doing Well

- No cough, wheeze, chest tightness, or shortness of breath during the day or night
- Can do usual activities

And, If a peak flow meter is used,

Peak flow: more than _____
 (80 percent or more of my best peak flow)

My best peak flow is: _____

Take these long-term-control medicines each day (Include an anti-inflammatory).

Medicine	How much to take	When to take it
_____	_____	_____

Identify and avoid and control the things that make your asthma worse, like (list here):

Before exercise, if prescribed, take: ☐ 2 or ☐ 4 puffs _____ 5 to 60 minutes before exercise

YELLOW ZONE

Asthma Is Getting Worse

- Cough, wheeze, chest tightness, or shortness of breath, or
- Waking at night due to asthma, or
- Can do some, but not all, usual activities

-Or-

Peak flow: _____ to _____
 (50 to 79 percent of my best peak flow)



Add: quick-relief medicine—and keep taking your GREEN ZONE medicine.

_____ ☐ 2 or ☐ 4 puffs, every 20 minutes for up to 1 hour
 (short-acting beta₂-agonist) ☐ Nebulizer, once

If applicable, remove yourself from the thing that made your asthma worse.

If your symptoms (and peak flow, if used) return to GREEN ZONE after 1 hour of above treatment:

- ☐ Continue monitoring to be sure you stay in the green zone.



If your symptoms (and peak flow, if used) do not return to GREEN ZONE after 1 hour of above treatment:

- ☐ Take: _____ ☐ 2 or ☐ 4 puffs or ☐ Nebulizer
 (short-acting beta₂-agonist)

- ☐ Add: _____ mg per day. For _____ (3–10) days
 (oral corticosteroid)

- ☐ Call the doctor _____ ☐ before/ ☐ within _____ hours after taking the oral corticosteroid.
 (phone)

RED ZONE

Medical Alert!

- Very short of breath, or
- Quick-relief medicines have not helped, or
- Cannot do usual activities, or
- Symptoms are same or get worse after 24 hours in Yellow Zone.

-Or-

Peak flow: less than _____
 (50 percent of my best peak flow)

Take this medicine:

- ☐ _____ ☐ 4 or ☐ 6 puffs or ☐ Nebulizer
 (short-acting beta₂-agonist)

- ☐ _____ mg
 (oral corticosteroid)

Then call your doctor NOW. Go to the hospital or call an ambulance if

- You are still in the red zone after 15 minutes AND
- You have not reached your doctor.

DANGER SIGNS ■ Trouble walking and talking due to shortness of breath
 ■ Lips or fingernails are blue



■ Take ☐ 4 or ☐ 6 puffs of your quick-relief medicine AND
 ■ Go to the hospital or call for an ambulance _____ NOW!
 (phone)

FIGURE 78.5 Asthma self-management guideline. (From National Heart Lung Blood Institute, National Asthma Education and Prevention Program. Expert panel report 3: guidelines for the diagnosis and management of asthma. Summary report 2007, NIH Publication No. 05-5251, Bethesda, MD, 2007, U.S. Department of Health and Human Services, 119, <http://www.nhlbi.nih.gov/guidelines/asthma/asthgdh.pdf>.)

Nelson's Essentials of Pediatrics, 2023; National Heart, Lung, and Blood Institute

Case 3

A female born at 38wks gestation from a primigravid mom via SVD is noted to have darkly stained and foul-smelling amniotic fluid upon delivery.

You note low tone and cyanosis of the extremities. Apgar scores are 5 and 6 at 1 and 5 minutes, respectively.

Neonatal resuscitation is performed, and the infant is admitted to the NICU.

Case 3: Physical Exam

General: nasal flaring, grunting, increased A-P diameter to the chest, intercostal retractions

Vitals: T 98.8F, Resp 70, HR 165, O2 sat 93% w/O2 by nasal cannula

Cardio: normal S1 and S2 without MRG

Resp: Diffuse rhonchi, no wheezes or rales

Case 3: What is the diagnosis?

- A. Meconium Aspiration Syndrome
- B. Transient Tachypnea of the Newborn
- C. Pneumonia
- D. Hydrops Fetalis
- E. Neonatal Respiratory Distress Syndrome

Case 3: What is the diagnosis?

- A. Meconium Aspiration Syndrome
- B. Transient Tachypnea of the Newborn
- C. Pneumonia
- D. Hydrops Fetalis
- E. Neonatal Respiratory Distress Syndrome

Meconium Aspiration Syndrome (MAS)

- Seen in 5% of newborns with meconium-stained amniotic fluid
- Aspiration → airway obstruction → respiratory distress
- X-ray: infiltrates, streaking, ↑A-P diameter, flattened diaphragms
- Meconium effects on the lungs:
 - Ball-valve effect
 - Chemical irritation
 - Predisposition for infection
 - Inhibition of production and function of surfactant
- Often seen with primary pulmonary hypertension of the newborn (↑pulmonary vascular resistance)

<https://www.merckmanuals.com/home/children-s-health-issues/lung-and-breathing-problems-in-newborns/meconium-aspiration-syndrome>

MAS Management

- Supportive measures + assisted ventilation
- Surfactant therapy
- +/- antibiotics
- Inhaled nitric oxide
- If the above fail
 - Extracorporeal membrane oxygenation (ECMO) – modified heart-lung apparatus

Differential Diagnosis: Newborn Respiratory Distress

- Preterm
 - Respiratory Distress Syndrome (“Hyaline Membrane Disease”) (RDS)
 - Apnea of prematurity
- Full-term and post-term
 - Primary Pulmonary Hypertension of the Neonate (PPHN)
 - Meconium Aspiration Pneumonia
- Preterm and full-term
 - Transient Tachypnea of the Newborn
 - Bacterial Sepsis (Group B *Streptococcus*)

Adapted from Table 61.1 in Nelson’s Essentials of Pediatrics (pathologies denoted as common)

Respiratory Distress: Lab Evaluation

- Chest X-Ray:
 - Parenchymal pattern determination, ?pneumothorax, cardiac morphology
- Arterial Blood Gas:
 - Acid-base disturbance, extent of hypercapnia/hypoxia
- Complete Blood Count:
 - H+H, white count, platelet count (?DIC)
- Blood Culture:
 - Recovery of offending microbe
- Blood Glucose:
 - Hypoglycemia, stress hyperglycemia
- Echocardiogram:
 - Interrogation of murmur or PPHN, cardiomegaly, structural heart disease

Adapted from Table 61.2 in Nelson's Essentials of Pediatrics

Session 2

Case 4

A 4-month-old infant presents to the ED with cough and runny nose for the past 3 days.

She was given APAP for tactile fever(felt hot).

The patient lives at home with her parents, her aunt and 9-month-old cousin.

PMHx: noncontributory

FmHx: HTN grandparents

Case 4

Vitals: HR 166, RR 48, Temp 100.2F, O2 Sat. 90% on room air

General: in mild distress with noticeable increased work of breathing

HEENT: clear nasal d/c with alar flaring, rest is normal

CV: Reg. rhythm, S1 and S2 present without S3, S4, MRG

Resp: supraclavicular retractions, diffuse wheeze and crackles heard throughout all lung fields

Case 4: What is the diagnosis?

- A. Asthma
- B. Pneumonia
- C. Bronchiolitis
- D. Allergic Rhinitis

Case 4: What is the diagnosis?

- A. Asthma
- B. Pneumonia
- C. Bronchiolitis**
- D. Allergic Rhinitis

Bronchiolitis: Etiology + Epidemiology

- Causes:
 - Respiratory syncytial virus (**RSV**)
 - Others: human metapneumovirus, parainfluenza, influenza, adenovirus, rhinovirus, coronavirus
- Usually ≤ 2 years of age (peak: 2-6 mos old)
- Increased risk: premature infants, chronic lung disease, congenital heart disease, neuromuscular weakness, immunodeficiency
- FDA 7/17/23: Beyfortus(nirsevimab) monoclonal antibody against RSV, IM single dose 0-24 mos old

Bronchiolitis: Natural History + Clinical Signs

- Course:
 - Incubation (4-6 days)
 - 3-7day progression to noisy breathing, wheezing, increased work of breathing (WOB)
 - Pulmonary dysfunction and bronchial hyperreactivity may persist for several years
- Signs: prolonged expiration, wheeze, nasal flaring, suprasternal retractions, air trapping + lung hyper-expansion
- Grunting and cyanosis = severe disease
- Young infants can present with apnea

Time Course

Source: Lancet
2017; 389: 211–24

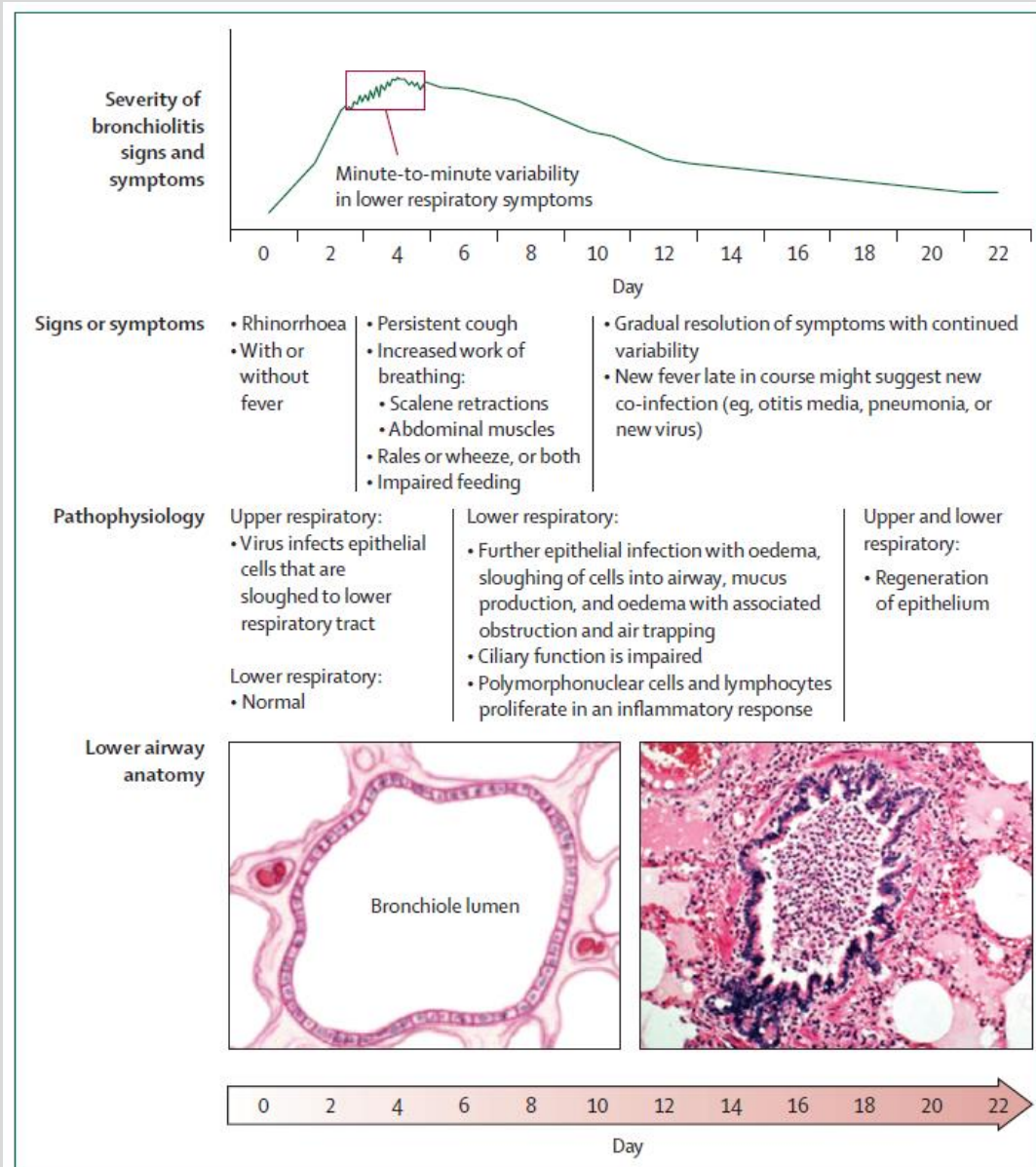


Figure 1: Typical clinical course and pathophysiology of viral bronchiolitis

Did this [CXR] need to be ordered?

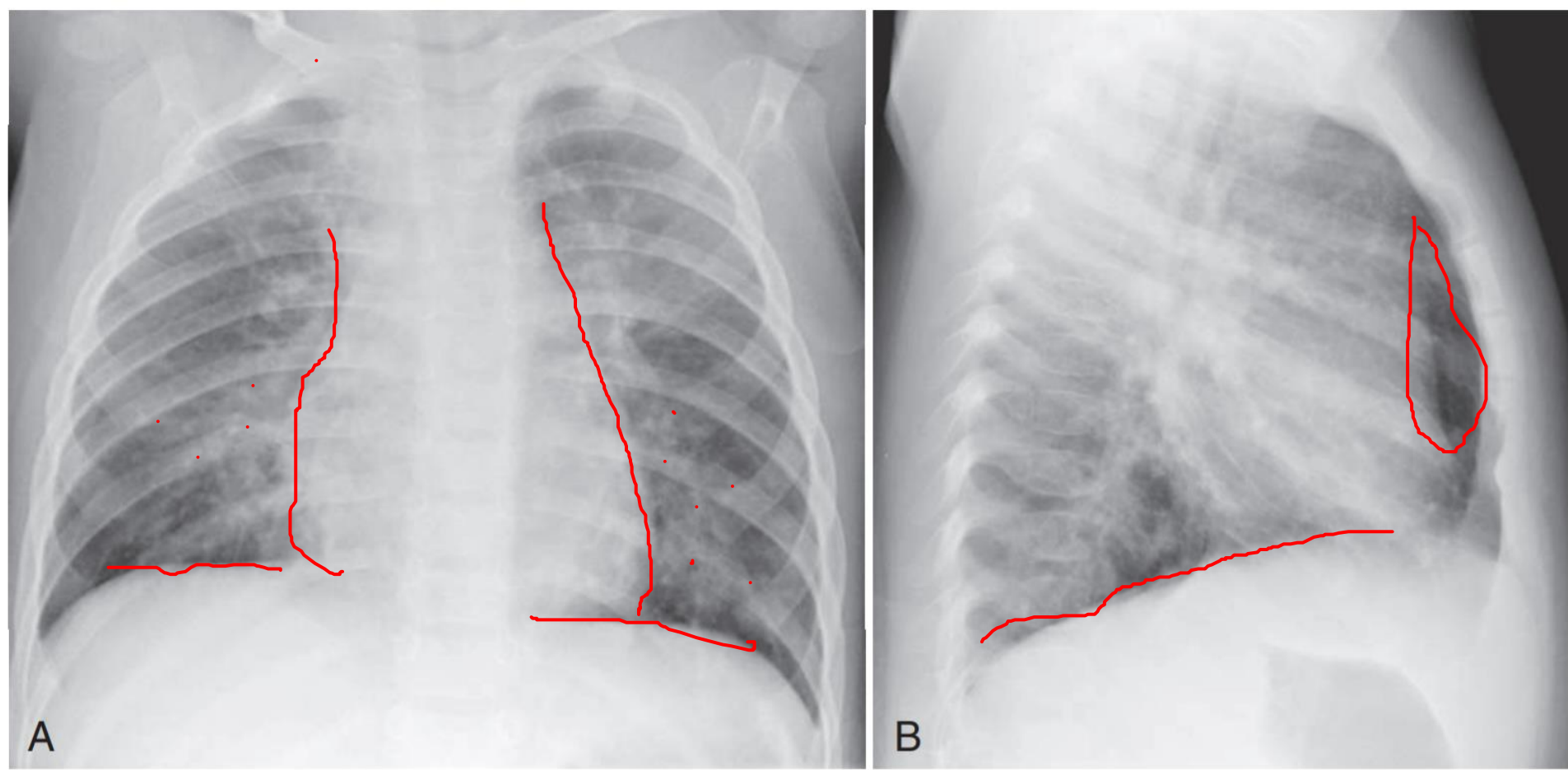


Image: Patricia Trinidad Chang and Laureen Sena

“Chest Imaging.” Pediatric Radiology: The Requisites, Chapter 2, 6-61. Figure 2.2

Case 4: Lab + Imaging Studies

- Monitoring with pulse oximetry
- Rapid antigen testing or polymerase chain reaction(PCR) to verify etiology: RSV, parainfluenza, influenza, adenovirus (nasal swab) can be done
- CXR: hyperinflation, flattened diaphragm
 - Increased density → pneumonia
 - Atelectasis

Differential Diagnosis: Why isn't it asthma?

- Differentiation from asthma: fever, age, lack of family history
- Other:

TABLE 78.1 Differential Diagnosis of Cough and Wheeze in Infants and Children

UPPER AIRWAY DISEASES	OBSTRUCTION INVOLVING LARGE AIRWAYS	OBSTRUCTION INVOLVING SMALL AIRWAYS	OTHER
Allergic rhinitis Sinusitis	Tracheal or bronchial foreign body Paradoxical vocal fold motion Vascular rings or laryngeal webs Enlarged lymph nodes or tumor	Viral bronchiolitis or obliterative bronchiolitis Cystic fibrosis Bronchopulmonary dysplasia (chronic lung disease of prematurity)	Recurrent cough not caused by asthma (infection, habit cough, postnasal drip) Aspiration from swallowing mechanism dysfunction or gastroesophageal reflux disease

From Guilbert TW, Lemanske RF, Jackson DJ. *Diagnosis of asthma in infants and children*. In: Adkinson NF Jr, Bochner BS, Burks AW, et al., eds. *Middleton's Allergy: Principles and Practice*. 8th ed. Philadelphia: Saunders; 2014:870.

Bronchiolitis: Treatment

- Supportive therapy
 - Antipyretic
 - Adequate hydration
 - Nasal suctioning
 - Oxygen administration as needed
- Usually avoided: antibiotics, bronchodilators, corticosteroids
- 1-2% mortality rate

Case 5

You receive a call from a colleague in the E.D. that your 9-year-old patient, whom you saw in the fall with a known history of asthma, has presented again. This time with fever, cough, and chills. You go to the E.D. to see him yourself.

Case 5: Physical Exam

General: ill-appearing, pale and diaphoretic with an audible wet cough

Vitals:

T: 102.5F, HR: 110, RR: 32, BP: 116/82

Cardio: S1 and S2 present with no MRG; capillary refill delayed at >2 sec

Resp: Use of accessory muscles to breathe. Bronchophony in the right lower lobe with dullness to percussion. BS heard in all fields with prolonged expiratory phase. Expiratory wheeze with crackles and rhonchi heard over the right lower lobe.

Case 5: Chest X-Ray



Nelson's Essentials of Pediatrics Figure 110.1

Case 5: What is the diagnosis?

- A. It's still Asthma
- B. Laryngitis
- C. Pneumothorax
- D. Pneumonia

Case 5: What is the diagnosis?

- A. It's still Asthma
- B. Laryngitis
- C. Pneumothorax
- D. Pneumonia

Pneumonia: Epidemiology

- Decreased incidence because of vaccine development:
 - Pertussis, Diphtheria, Measles, *H. flu* type b, *S. pneumo(pneumococcus)*, Influenza, Covid
 - Worldwide: kills 1 million children under age 5 annually
- Risk factors: gastroesophageal reflux, neuro impairment, immunocompromised, anatomical abnormalities, group homes, hospitalization

Pneumonia: Clinical Distinctions

Viral

- Cough, wheeze, stridor
- Less prominent fever
- Mucosal congestion, upper airway inflammation
- **CXR:** diffuse infiltrates and hyperinflation

Bacterial

- High fever, chills, cough, dyspnea, consolidation (auscultatory findings)
- Dullness to percussion → segmental infiltrates
- **CXR:** lobar consolidation, round pneumonia + pleural effusion (10-30% of cases)

Pediatric Pneumonia: Microbiology by Age Group

Age Group		Pathogen
≤1 month		Group B <i>Strep</i> , <i>E. coli</i> , Gram (-) bacilli, <i>S. pneumoniae</i>
1-3 mos	Febrile	RSV and other viruses
	Afebrile	<i>C. trachomatis</i> , <i>M. hominis</i> , <i>U. urealyticum</i> , cytomegalovirus(CMV), <i>B. pertussis</i>
3 months to 5 years		RSV + other viruses, <i>S. pneumo</i> , <i>H. influenzae</i> (type b, non-typeable),
5-18y		<i>Mycoplasma pneumoniae</i> , <i>S. pneumoniae</i> , <i>C. pneumoniae</i>
≥18y		<i>M. pneumoniae</i> , <i>S. pneumoniae</i> , <i>C. pneumoniae</i> , <i>H. influenzae</i> (type b, non-typeable), influenza, adenovirus

Pneumonia: Management

- Symptoms, physical exam, imaging, age will typically direct the care
- Etiology isolated/identified 40-80% of the time
 - More urgent to discern in hospitalized patients
- Empyema or pleural effusion: thoracentesis is diagnostic and therapeutic
- Surgical management: abscesses, pneumatocele, empyema
- **For this patient, pneumonia can trigger a new asthma exacerbation, and asthma can predispose a pneumonia**

Case 6

A previously healthy 2-year-old female is brought to the E.D. with a 2h history of fever and “harsh breathing while she inhales,” according to the mother. There are no known sick contacts or witnessed episodes of choking.

Per the parents’ wishes, the patient is unimmunized.

Adapted from:

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3380746/pdf/pch17307a.pdf>

Case 6: PE

General: hoarse cry, inspiratory stridor, suprasternal retractions, toxic appearing

Vitals: T 102 F, HR 150 bpm, RR 40 bpm, O2 sat 97% (room air)

Resp: High pitched noise over trachea on inspiration, lungs CTA bilaterally without wheezes, rales, or rhonchi

CV: normal S1 and S2 with no S3, S4, murmurs, rubs, or gallops

Case 6: Most likely diagnosis

- A. Epiglottitis
- B. Retropharyngeal abscess
- C. Laryngotracheobronchitis
- D. Peritonsillar abscess
- E. Bacterial tracheitis

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Case 6: CXR

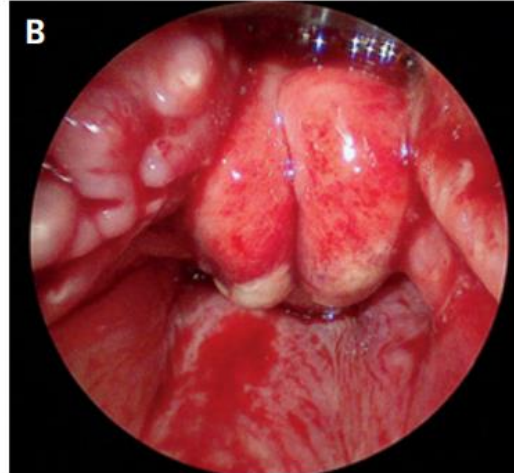


Epiglottitis

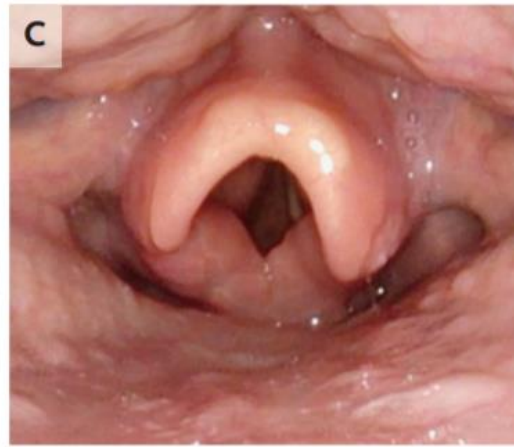
- Etiology: Group A *Streptococcus*, *S. aureus*, *H. influenzae* type b (unimmunized)
- Radiograph: “thumb sign”, inflamed epiglottis
- Dx: visualization when placing an endotracheal tube
- Management: calm quiet environment, secure the airway and treat with IV antibiotics/steroids

It can happen in adults, too:

N ENGL J MED 381;9 NEJM.ORG AUGUST 29, 2019



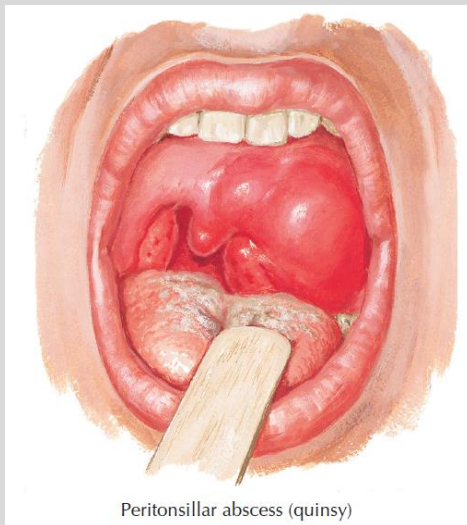
Laryngoscopy in the E.D.



After 5d IV abx +
14d of amoxicillin-
clavulanic acid
("Augmentin")

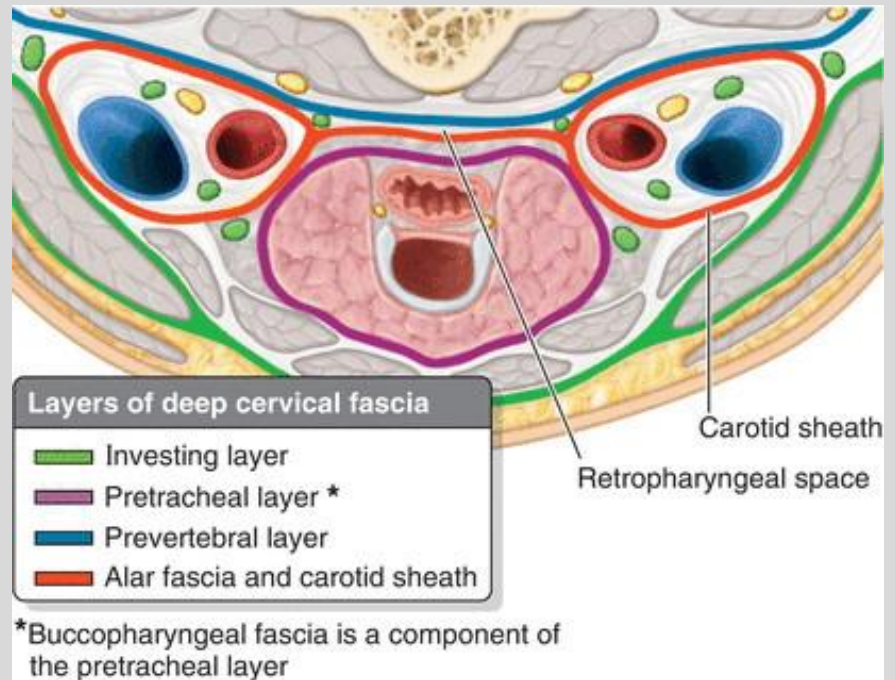
Other causes of Stridor

- Retropharyngeal abscess
- Foreign body aspiration
- Laryngotracheobronchitis (croup)
- Peritonsillar abscess
- Airway hemangioma
- Subglottic stenosis
- Vocal cord paralysis



Peritonsillar abscess (quinsy)

Netter's Pediatrics Fig. 35-4



Moore's Clinically Oriented Anatomy 8e Fig. 9.6

Summary Slide

- **Cystic Fibrosis:** multiple mutations → sweat Cl testing, multiorgan involvement, chronic condition, improving lifespan
- **Asthma:** reversible, episodic; may need *maintenance* therapy; many triggers
- **Meconium Aspiration Syndrome:** multiple effects on lungs; different from other newborn resp. distress causes
- **Bronchiolitis:** usually RSV, peak 2-6 mos. of age
- **Pneumonia:** age group helps predict etiology, guides treatment; viral vs. bacterial clinical and x-ray patterns
- **Epiglottitis:** often vaccine-preventable, inflammation of epiglottis on lat. Neck x-ray, clinical signs and PE differentiate from other causes of respiratory distress

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>